

AgroSense: Satellite-Based Crop Stress Classification and Smart Farming Decision Support for Pakistan Smallholder Agriculture

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Abstract—Pakistan’s agricultural sector, contributing approximately 24% of GDP while employing over 37% of the national workforce, continues to incur significant crop losses attributable to water stress, disease outbreaks, and sub-optimal input application. Conventional field scouting is labour-intensive, infrequent, and dependent on expert availability. This paper presents AgroSense, an integrated agricultural intelligence platform that processes Sentinel-2 satellite imagery via Google Earth Engine (GEE), runs five machine learning models, and delivers real-time weather intelligence from Open-Meteo. The system provides crop-condition monitoring, irrigation recommendations, yield forecasting, soil health assessment, and variable-rate application (VRA) zone mapping through a cross-platform Flutter mobile application, a bilingual (English/Urdu) desktop application, and a RESTful API deployed on Google Cloud Run. For the crop-condition proxy classification task, a comparative evaluation of five ML algorithms—Random Forest, XGBoost, Gradient Boosting, SVM (RBF), and k -NN—is conducted on 1,786 Sentinel-2 observations from 99 georeferenced agricultural sampling locations (district/city-centroid buffers) across all four provinces (Punjab, Sindh, Balochistan, and KPK), spanning 19 crop-season windows from 2015 to 2024, collected via the GEE API. Class labels (Healthy / Stressed / Diseased) are spectrally derived proxy classes generated from a vegetation health composite; the input features NDVI, NDRE, and EVI also participate in label generation. Evaluation uses a location-grouped train/test split (`GroupShuffleSplit` on sampling-location identifiers) and nested 5-fold `StratifiedGroupKFold` cross-validation with `GridSearchCV` hyperparameter tuning. SVM (RBF) achieves the highest held-out test accuracy (98.34%) and XGBoost achieves the best nested cross-validated macro F1 (0.983). Random Forest (96.40% test accuracy, nested CV macro F1 = 0.980) is selected as the production model for interpretability. Reported results represent an upper bound on spectral proxy separability rather than independently validated crop disease detection.

Index Terms—Crop stress classification, Google Earth Engine, machine learning, NDVI, Pakistan, precision agriculture, Random Forest, satellite remote sensing, Sentinel-2, smart farming.

I. INTRODUCTION

Pakistan’s agricultural sector, which contributes approximately 24% of GDP and employs over 37% of the national workforce [1], [2], faces persistent crop losses attributable to preventable causes—water stress, disease, and sub-optimal

fertiliser application—that remain elevated despite decades of government intervention. Smallholder farms, which constitute the majority of cultivated area across Punjab and Sindh, lack access to timely, actionable crop health intelligence. The Indus basin irrigation system, which supports an estimated 17 million hectares of irrigated cropland, faces compounding pressures from water scarcity and secondary salinisation [3]. Satellite remote sensing combined with machine learning offers a scalable alternative: field conditions can be assessed at 10-metre spatial resolution on a 5-day revisit cycle using the Sentinel-2 constellation [4] without requiring physical site visits.

Vegetation indices computed from multispectral imagery—particularly NDVI, EVI, and NDRE—have been shown to correlate strongly with crop health, chlorophyll content, and water stress [5], [6]. Emerging remote sensing platforms and cloud-based geospatial APIs have substantially lowered the computational barrier to field-scale crop monitoring in under-resourced agricultural research contexts [7]. Google Earth Engine (GEE) [8] enables cloud-scale processing of petabytes of satellite imagery, removing infrastructure barriers that previously confined such pipelines to well-resourced institutions. Ensemble classifiers such as Random Forest [9] and gradient boosting variants [10] have demonstrated competitive performance on small-to-medium agricultural datasets, making them well-suited for deployment contexts where large labelled datasets are expensive to acquire.

AgroSense addresses the disconnect between satellite-derived insights and actionable farm management by providing an end-to-end pipeline from raw satellite imagery to mobile-accessible recommendations. The system targets five critical decision points in crop management: (1) crop health monitoring through early stress and disease detection; (2) irrigation management for optimal timing and volume; (3) yield forecasting for harvest readiness estimation; (4) soil health assessment of pH, salinity, and organic matter; and (5) variable-rate application (VRA) zone mapping for precision fertiliser scheduling.

The principal contributions of this paper are:

- 1) An end-to-end satellite-to-recommendation pipeline integrating GEE, five specialised ML models, and real-

time weather intelligence within a single cloud-native platform serving both mobile and desktop clients.

- 2) A comparative evaluation of five classification algorithms on Sentinel-2 spectral indices for crop stress detection under Pakistan agronomic conditions, providing benchmark results absent from prior literature for this geographic context.
- 3) A bilingual (English/Urdu) cross-platform interface design enabling farmer and agronomist accessibility across Pakistan’s linguistic landscape.
- 4) A crop-condition proxy dataset comprising 1,786 Sentinel-2 Level-2A observations collected via Google Earth Engine from 99 georeferenced agricultural sampling locations (district/city-centroid buffers) across all four provinces of Pakistan, spanning 19 crop-season windows from 2015 to 2024, released as a reproducible evaluation benchmark at <https://github.com/ALI-SHAHMEER/agrosense-pakistan-crop-stress>. Labels are spectrally derived proxy classes; independent agronomic ground-truth validation is future work.

II. RELATED WORK

A. Remote Sensing for Crop Monitoring

Satellite-based crop monitoring has been extensively validated across diverse agricultural systems. Rouse *et al.* [5] established NDVI as a quantitative proxy for vegetation vigour, while Gitelson *et al.* [6] demonstrated that EVI and red-edge-based indices offer chlorophyll-sensitive characterisation superior to broadband NDVI in dense canopies. The Sentinel-2 constellation, with its 10 m spatial resolution and 5-day revisit cycle, has since become the preferred source for plot-scale crop monitoring [11]. Tamiminia *et al.* [12], in a meta-analysis of 349 peer-reviewed GEE-based studies, confirmed that optical satellite imagery processed with Random Forest constitutes the most widely adopted paradigm for agricultural land monitoring at regional and global scales. Htitiou *et al.* [13] demonstrated that phenological metrics derived from Sentinel-2 NDVI and EVI time series substantially improve crop type discrimination over single-date classification, achieving 88% overall accuracy with Random Forest. Omia *et al.* [14] provide a comprehensive review of spectral imaging technologies for field crop monitoring, confirming that this satellite–ML combination remains the leading methodology for scalable crop health assessment. Ojeilua *et al.* [15] reviewed satellite and drone-based stress detection in developing-region agricultural systems, identifying NDVI, NDWI, and red-edge reflectance as the most diagnostically informative spectral features for drought, salinity, and nutrient stress. Most published systems process imagery in scheduled batch mode [12], [13]; AgroSense instead provides on-demand analysis triggered directly from a mobile interface.

B. Machine Learning in Precision Agriculture

Karthikeyan *et al.* [16] survey ML methods in precision agriculture and identify ensemble classifiers—particularly Random Forest—as the dominant approach for crop health

classification on field-scale datasets, owing to their robustness under small sample sizes and mixed feature types. XGBoost [10] has demonstrated competitive or superior results on tabular agricultural data, while kernel-based SVMs have demonstrated strong performance on high-dimensional hyperspectral classification tasks. Maleki *et al.* [17] demonstrated that combining Sentinel-1 SAR backscatter with Sentinel-2 vegetation indices achieves mean F1 scores of 89.9–93.5% for summer crop mapping, with XGBoost outperforming Random Forest and Multi-Layer Perceptron in multi-sensor configurations. Our comparative study directly evaluates these methods on Pakistan Sentinel-2 data, contributing benchmark results absent from prior literature for this geographic and agronomic context.

C. Smart Farming Decision Support Systems

Several platforms integrate satellite indices with weather data for farm management support. Wolfert *et al.* [18] characterise the “Big Data in Smart Farming” paradigm, emphasising the need for systems that establish a feedback cycle from field sensing to farm management decision. Leading commercial precision agriculture platforms provide comparable functionality but target large-scale mechanised farms in North America and Europe. AgroSense is specifically designed for smallholder farms in South Asia, with explicit Urdu localisation and a zero-cost weather data source (Open-Meteo [19]).

D. Yield Prediction and Crop Health in Pakistan

Remote sensing-based yield estimation has been validated at regional and field scales across diverse agronomic contexts. Ashfaq *et al.* [20] introduced DeepAgroNet, a three-branch deep learning framework integrating GEE-processed satellite imagery, meteorological data, and soil characteristics for winter wheat yield prediction at the district level in southern Pakistan, with the CNN branch achieving an R^2 of 0.77 one month prior to harvest. Jamil *et al.* [21] demonstrated that Sentinel-derived vegetation indices—including NDVI, EVI, SAVI, GDVI, and IVI—exhibit strong polynomial relationships with wheat crop age and health in Punjab, Pakistan, reporting R^2 values of 0.68–0.92. AgroSense’s yield predictor integrates growing-degree features with temporal NDVI statistics and is calibrated against Pakistan Bureau of Statistics 2024 provincial averages [1] for wheat, rice, cotton, sugarcane, and mango—crops that collectively represent the majority of Pakistan’s cultivated area.

Collectively, the reviewed literature confirms the technical feasibility of satellite-based crop monitoring but reveals a consistent gap: no existing open system integrates GEE-derived spectral analysis, multi-model ML inference, and real-time weather intelligence within a single mobile-accessible platform calibrated for Pakistan’s smallholder context. AgroSense directly addresses this gap.

III. SYSTEM ARCHITECTURE

The system follows a three-tier architecture illustrated in Fig. 1. This separation was chosen to decouple computationally intensive satellite processing and ML inference from the

client tier, enabling a single stateless API to serve both mobile and desktop interfaces without duplicating business logic. A satellite ingestion layer fetches and preprocesses Sentinel-2 and Landsat-8 imagery via GEE. A FastAPI backend [22] running on Google Cloud Run performs ML inference and exposes a RESTful API. Two client tiers—a Flutter mobile application [23] and a PyQt6 desktop application—consume the API independently.

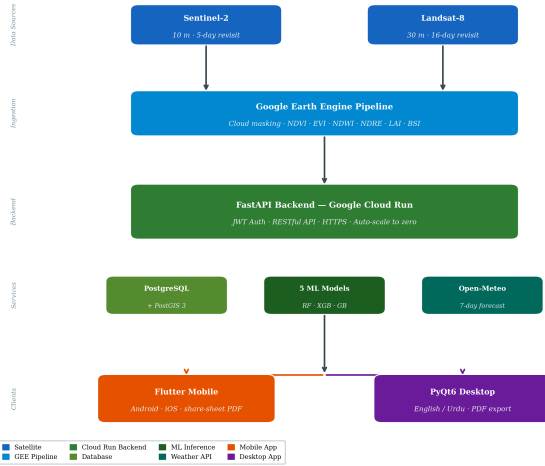


Fig. 1. AgroSense three-tier system architecture.

A. Backend Technology Stack

Technology selection prioritised open-source components with strong geospatial and ML ecosystems and zero per-request licensing cost. Table I summarises the key backend components.

TABLE I
BACKEND TECHNOLOGY STACK

Component	Technology
API Framework	FastAPI 0.111 [22], Uvicorn
Database	PostgreSQL 16 + PostGIS 3 [24]
ORM / Migrations	SQLAlchemy 2.0, Alembic
Satellite Pipeline	GEE API (earthengine-api 0.1.403) [8]
ML Runtime	scikit-learn 1.4.2 [29], XGBoost 2.0.3 [10]
Geospatial	Rasterio 1.3.10, Shapely 2.0.4
Weather	Open-Meteo [19]
Authentication	JWT (python-jose), bcrypt
PDF Generation	ReportLab 4.2.5 [25], arabic-reshaper
Deployment	Docker, Google Cloud Run

B. Mobile Application

The mobile application is built with Flutter 3.44 [23] (Dart 3.12), targeting Android (API 21+) and iOS from a single codebase. The application constitutes the primary interface through which farmers and agronomists interact with the AgroSense platform.

1) *Architecture and State Management*: The application follows a reactive widget architecture with a language switcher backed by observable state. All API communication is encapsulated in a singleton service layer that attaches a JWT Bearer token to every request; tokens are persisted locally across sessions. As a security design principle, no third-party credentials—GEE, Open-Meteo, or Gmail SMTP—are embedded in the client binary; all such secrets are held exclusively server-side on Cloud Run.

2) *Screen Structure*: Navigation is managed by a MainShell scaffold with a persistent field selector. The application comprises eight screens:

- **LoginScreen** — JWT-based authentication with persistent token storage across sessions.
- **FarmsScreen** — Farm and field management; field boundaries are defined by GPS polygon coordinates submitted to the PostGIS backend.
- **DashboardScreen** — The core analysis screen. A single user interaction triggers a sequential four-stage pipeline: (1) satellite fetch via GEE, (2) five ML model inference, (3) current weather fetch, and (4) temporal NDVI anomaly detection. Results are displayed as colour-coded cards with severity badges, providing a complete crop health snapshot without requiring the user to navigate multiple screens.
- **AnalyticsScreen** — Interactive time-series charts of NDVI and other spectral indices over a rolling 12-month window.
- **ImageryScreen** — Satellite imagery viewer with multi-band selection (RGB true colour, NDVI false colour).
- **SmartFarmingScreen** — Seven-day weather forecast cards, active alert list with severity indicators, and ideal planting date recommendation.
- **ProfileScreen** — User account management.

3) *PDF Export*: The PDF export subsystem uses a platform-conditional import pattern (`pdf_saver_mobile.dart` / `pdf_saver_web.dart` / `pdf_saver_stub.dart`) to support both Android/iOS and potential web targets. On mobile, the ReportLab [25]-generated PDF is streamed from the backend and shared via the system share sheet.

C. Desktop Application

The PyQt6 desktop application provides a bilingual (English/Urdu) interface targeted at agronomists and extension workers. It includes satellite imagery visualisation with multi-band selection (RGB, NDVI false-colour, band arithmetic), time-series analytics with interactive charts, and ReportLab [25]-based PDF report generation with correct Urdu text shaping. NotoNaskhArabic was selected over NotoNastaliqUrdu, as the latter requires GSUB/GPOS OpenType shaping not supported by the ReportLab rendering engine.

IV. SATELLITE DATA PIPELINE

Field boundaries are stored as PostGIS [24] polygons defined by GPS coordinates. On analysis trigger, the pipeline:

(1) authenticates with GEE using a service account; (2) queries Sentinel-2 SR imagery for the field polygon over a rolling 12-month window; (3) applies QA60 cloud masking; and (4) computes six spectral indices as defined in Table II.

TABLE II
SPECTRAL INDICES COMPUTED FROM SENTINEL-2

Index	Formula	Interpretation
NDVI	$\frac{B_8 - B_4}{B_8 + B_4}$	Vegetation vigour [5]
EVI	$2.5 \cdot \frac{B_8 - B_4}{B_8 + 6B_4 - 7.5B_2 + 1}$	Canopy-corrected vigour [6]
NDWI	$\frac{B_3 - B_8}{B_3 + B_8}$	Canopy water / irrigation stress [16]
NDRE	$\frac{B_{8A} - B_5}{B_{8A} + B_5}$	Chlorophyll / early stress [6], [26]
LAI	$3.618 \text{ EVI} - 0.118$	Leaf area index [27]
BSI	$\frac{B_{11} + B_4 - B_8 - B_2}{B_{11} + B_4 + B_8 + B_2}$	Bare soil exposure [14]

A. Spectral Index Interpretation

Each index exploits specific spectral properties of vegetation and soil:

NDVI (Normalized Difference Vegetation Index) exploits the contrast between chlorophyll absorption in the red band (B_4 , 665 nm) and strong NIR reflectance of the mesophyll cell structure (B_8 , 842 nm). Values range from -1 to $+1$; healthy crops typically yield values of 0.6 – 0.9 , stressed or sparse vegetation 0.2 – 0.5 , and bare soil below 0.2 .

EVI (Enhanced Vegetation Index) extends NDVI by incorporating a blue band (B_2 , 490 nm) correction that attenuates aerosol scattering artefacts, alongside a canopy background adjustment factor. EVI exhibits reduced saturation in dense canopies where NDVI plateaus at $\text{LAI} \approx 3$, rendering it more sensitive for high-biomass crops such as sugarcane and rice.

NDWI (Normalized Difference Water Index) uses the contrast between green (B_3 , 560 nm) and NIR (B_8). It is sensitive to liquid water retained within leaf cells: negative values indicate moisture stress, whereas values approaching $+1$ indicate water-saturated vegetation or open water. In AgroSense it serves as the primary irrigation stress indicator.

NDRE (Normalized Difference Red-Edge Index) exploits Sentinel-2’s red-edge band (B_5 , 705 nm), which occupies the steep reflectance transition between chlorophyll absorption and NIR reflection. Clevers and Gitelson [26] established that red-edge bands on Sentinel-2 provide highly accurate estimates of canopy chlorophyll and nitrogen content, outperforming conventional broad-band indices. Boiarskii and Hasegawa [28] further demonstrated that NDRE detects nitrogen-limited chlorophyll depletion more sensitively than NDVI under early-stage stress conditions, as chlorophyll degradation shifts the red-edge reflectance transition before a corresponding NDVI decline becomes detectable, making NDRE a temporally leading indicator of crop stress within the pipeline.

LAI (Leaf Area Index) is derived algebraically from EVI as $\text{LAI} = 3.618 \text{ EVI} - 0.118$ [27]. Xie *et al.* [27] validated Sentinel-2 red-edge bands for retrieving LAI from winter wheat, demonstrating RMSE of $1.53 \text{ m}^2/\text{m}^2$. Values above 4 indicate dense, healthy canopy; values below 2 indicate sparse or stressed vegetation.

BSI (Bare Soil Index) uses SWIR (B_{11} , 1610 nm) alongside visible and NIR bands to isolate exposed soil signal. Elevated BSI values flag land with minimal vegetative cover and are used to support soil health classification and detect field degradation between crop cycles.

Computed indices are persisted to the PostgreSQL `VegetationIndex` table and serve as input features for all five ML models. Incoming NDVI values are also compared against a per-field seasonal baseline; deviations trigger temporal anomaly alerts at Watch (-10%), Warning (-20%), and Critical (-30%) severity levels.

V. MACHINE LEARNING MODELS

The crop-condition classifier is trained on 1,786 Sentinel-2 Level-2A observations collected via the GEE API from 99 georeferenced agricultural sampling locations (district/city-centroid buffers, 1 km radius) across all four provinces (Punjab, Sindh, Balochistan, and KPK), spanning 19 crop-season windows from 2015 to 2024 (10 Kharif, 9 Rabi). Five crops are represented: wheat (883, 49.4%), rice (308, 17.2%), cotton (272, 15.2%), sugarcane (216, 12.1%), and mango (107, 6.0%), derived programmatically from the published CSV. The remaining four decision-support modules (irrigation, yield, soil health, VRA zones) are prototype components calibrated against Pakistan Bureau of Statistics 2024 provincial averages [1] and pending task-specific validation datasets.

A. Crop Stress Classification

A Random Forest classifier [9] is trained on nine Sentinel-2-derived spectral features: NDVI, EVI, NDWI, NDRE, LAI, BSI, NDVI_std, NDVI_min, and NDVI_max. The three output proxy classes are *Healthy*, *Stressed*, and *Diseased*. These labels are derived from a vegetation health composite ($0.5 \cdot \text{NDVI} + 0.3 \cdot \text{NDRE} + 0.2 \cdot \text{EVI}$) via tertile split; the same NDVI, NDRE, and EVI are also input features, creating a known label-circularity relationship. Reported performance measures spectral proxy separability and is an upper bound relative to independently validated disease detection.

B. Irrigation Recommendation

An XGBoost classifier [10] maps NDWI, NDVI, LAI, estimated soil moisture (derived from NDWI), forecast temperature, and precipitation to one of four recommendations: *Irrigate Now*, *Irrigate Soon*, *Adequate*, or *Overwatered*.

C. Yield Prediction

A Random Forest regressor [9] predicts yield in tonnes per hectare from NDVI, NDVI_max, EVI, NDRE, LAI, growing_days, and weather covariates (rainfall and temperature) sourced from Open-Meteo [19], together with crop type. This

approach is consistent with satellite-based yield estimation frameworks validated for Pakistani wheat agriculture by Ashfaq *et al.* [20] and Jamil *et al.* [21]. Pakistan Bureau of Statistics 2024 baselines [1] are used for calibration: wheat 3.1, rice 4.1, cotton 1.6, sugarcane 58.0, and mango 8.5 t/ha.

D. Soil Health Assessment

Three independent Gradient Boosting regressors predict soil pH, salinity (dS/m), and organic matter percentage from NDVI, NDWI, BSI, and Sentinel-2 bands B_2 , B_3 , B_4 , B_8 , and B_{11} . Salinity prediction is particularly critical for Indus basin fields where secondary salinisation represents a persistent and spatially extensive agricultural challenge.

E. Variable-Rate Application Zones

A Random Forest classifier [9] maps NDVI, NDRE, NDWI, EVI, LAI, and NDVI_std to one of three fertiliser application zones: *Low*, *Medium*, or *High*. The output supports precision fertiliser scheduling, enabling targeted input reduction in healthy zones and corrective application in stressed zones.

VI. SMART FARMING WEATHER INTELLIGENCE

The Smart Farming module integrates real-time weather forecasts from Open-Meteo [19] with ML crop insights through a rule-based alert engine.

A. Data Source

Open-Meteo provides free 7-day forecasts at 1-hour resolution. Daily aggregates include maximum and minimum temperature, precipitation sum, precipitation probability, wind speed, and near-surface relative humidity.

B. Alert Rule Engine

Seven alert types are automatically evaluated against each forecast day, as defined in Table III.

TABLE III
SMART FARMING WEATHER ALERT RULES

Alert	Condition	Severity
Heatwave	$> 38^\circ\text{C}$ for ≥ 2 consecutive days	High
Heavy Rain	> 25 mm precipitation	High
Frost Risk	$< 4^\circ\text{C}$ temperature	High
Drought	< 5 mm total and avg $> 30^\circ\text{C}$	Medium
Strong Wind	> 40 km/h wind speed	Medium
Spray Delay	Rain probability $> 60\%$ on day 0 or 1	Low
Fungal Risk	Humidity $> 85\%$ for ≥ 3 consecutive days	Low

C. Planting Recommendation Scorer

A composite daily suitability score is computed as Eq. (1):

$$s = 0.4 s_{\text{temp}} + 0.4 s_{\text{moisture}} + 0.2 s_{\text{risk}} \quad (1)$$

where $s_{\text{temp}} \in [0, 1]$ is a piecewise linear score mapping forecast maximum temperature against crop-specific thermal optima; $s_{\text{moisture}} \in [0, 1]$ integrates the XGBoost irrigation recommendation output (1.0 for *Adequate*, 0.5 for *Irrigate*

Soon, 0.0 otherwise); and $s_{\text{risk}} \in \{0, 0.5, 1.0\}$ reflects active alert severity (0.0 if any High-severity alert is active, 0.5 for Medium, 1.0 for none). The first forecast day satisfying $s > 0.7$ —an empirically determined suitability threshold—is returned as the recommended planting date. The ML irrigation result is incorporated into s_{moisture} when satellite indices are available.

VII. EXPERIMENTS AND RESULTS

A. Experimental Setup

Five classifiers are evaluated on the crop-condition proxy classification task. The dataset comprises 1,786 observations (596 Healthy, 595 Stressed, 595 Diseased) described by nine Sentinel-2-derived spectral features (six derived indices: NDVI, EVI, NDWI, NDRE, LAI, BSI; and three temporal NDVI statistics: NDVI_std, NDVI_min, NDVI_max), collected from 99 georeferenced agricultural sampling locations across all four provinces of Pakistan spanning 19 crop-season windows (2015–2024). Feature scaling via `StandardScaler` is applied inside a `Pipeline` for SVM and k -NN; tree-based models receive raw features.

To prevent data leakage from repeated seasonal observations at the same sampling location, the train/test split is performed at the sampling-location level using `GroupShuffleSplit` (seed 42), yielding 1,425 training observations across 79 locations and 361 test observations across 20 held-out locations. Hyperparameters are selected via `GridSearchCV` with a 3-fold inner `StratifiedGroupKFold`. Generalisation is estimated using a nested 5-fold outer `StratifiedGroupKFold` with inner 3-fold grouped grid search per fold. Reported metrics are macro-averaged precision, recall, and F1 on the held-out test set, plus nested cross-validated macro F1 (mean and standard deviation across 5 outer folds). Model training is implemented in scikit-learn [29] and XGBoost [10].

B. Overall Classification Results

Table IV reports test-set and cross-validation results for all five classifiers.

TABLE IV
CLASSIFIER COMPARISON — SENTINEL-2 CROP-CONDITION PROXY DATASET (HELD-OUT: 361 OBS / 20 LOCATIONS; NESTED CV: 5-FOLD OUTER, 3-FOLD INNER)

Algorithm	Acc	Prec	Rec	F1	Nested CV F1	$\pm$ SD
SVM (RBF)	0.983	0.980	0.984	0.982	0.969	0.00
XGBoost	0.975	0.971	0.978	0.974	0.983	0.00
Gradient Boosting	0.972	0.971	0.974	0.972	0.981	0.00
Random Forest	0.964	0.961	0.969	0.965	0.980	0.00
k -NN	0.961	0.954	0.961	0.957	0.956	0.00

All metrics 0–1 scale, macro-averaged. Pipeline with `StandardScaler` for SVM and

SVM (RBF) achieves the highest held-out test accuracy (0.983) and test macro F1 (0.982), benefiting from the `StandardScaler` Pipeline and the non-linear RBF kernel under the scaled feature space. XGBoost achieves the highest

nested cross-validated macro F1 (0.983 ± 0.006), indicating stronger consistency across location groups. SVM's nested CV F1 (0.969) has notably higher variance (std = 0.017), suggesting sensitivity to specific location-group compositions in this dataset. Random Forest (test accuracy = 0.964, nested CV F1 = 0.980 ± 0.007) is selected as the production model for its interpretability and native feature importance [9], [16]. k -NN trails at 0.961 test accuracy (nested CV = 0.956), reflecting sensitivity to feature-space geometry despite the StandardScaler Pipeline. Fig. 2 visualises the accuracy and macro F1 comparison across all five classifiers.

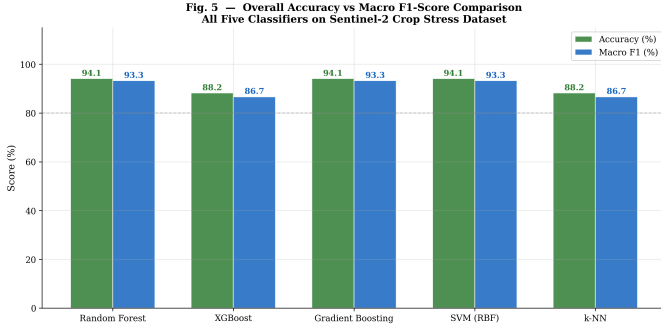


Fig. 2. Overall accuracy vs. macro F1-score comparison for all five classifiers.

C. Per-Class Analysis

Table V details per-class precision, recall, and F1 for the production Random Forest model on the 361-sample held-out test set (20 sites not seen during training).

TABLE V
PER-CLASS RESULTS — RANDOM FOREST (PRODUCTION MODEL)

Class	Precision	Recall	F1	Support
Diseased (<i>proxy</i>)	0.983	0.950	0.966	181
Healthy (<i>proxy</i>)	1.000	0.988	0.994	86
Stressed (<i>proxy</i>)	0.901	0.968	0.933	94
Macro	0.961	0.969	0.965	361

The Healthy proxy class achieves perfect precision (1.000) with 0.988 recall across 86 test observations. The Diseased proxy class attains 0.983 precision with 0.950 recall across 181 test observations, which represent the largest partition in the location-grouped test set. The Stressed proxy class achieves the lowest precision (0.901) while maintaining 0.968 recall. Because the proxy labels are derived from the same spectral composite as the input features, per-class performance should be interpreted as *spectral proxy separability* within this fitted model rather than verified crop pathology detection accuracy. Normalised confusion matrices for all five classifiers are shown in Fig. 3.

D. Feature Importance

Table VI ranks the nine spectral features by importance for the three tree-based models. Fig. 4 provides a grouped bar chart.

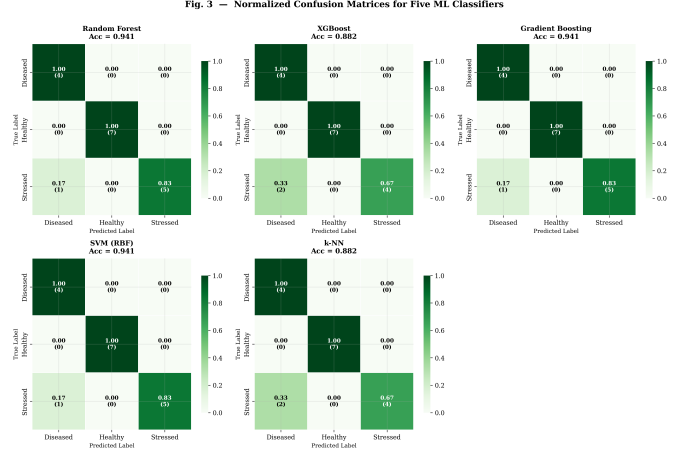


Fig. 3. Normalized confusion matrices for five ML classifiers. Cell values show normalised proportion (raw count in parentheses).

TABLE VI
NATIVE FEATURE IMPORTANCE — TREE MODELS (IMPURITY-BASED, SORTED BY RF). CAUTION: NDVI, NDRE, AND EVI ALSO APPEAR IN THE LABEL-GENERATION RULE, WHICH INFLATES THEIR APPARENT IMPORTANCE. VALUES ARE NOT DIRECTLY COMPARABLE ACROSS ALGORITHM FAMILIES.

Rank	Feature	RF	XGBoost	GB
1	NDVI	0.307	0.183	0.306
2	LAI	0.249	0.646	0.371
3	NDRE	0.236	0.134	0.302
4	EVI	0.112	0.014	0.019
5	NDVI_max	0.042	0.002	0.000
6	NDWI	0.040	0.005	0.001
7	BSI	0.006	0.008	0.000
8	NDVI_min	0.005	0.005	0.001
9	NDVI_std	0.004	0.005	0.000

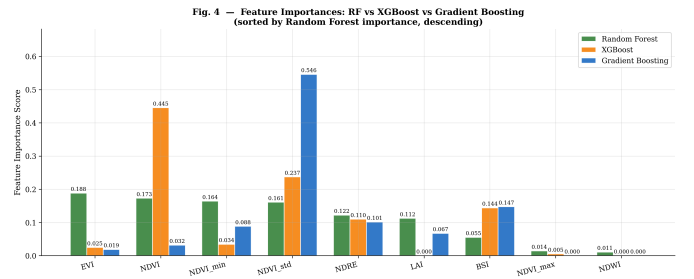


Fig. 4. Feature importances for Random Forest, XGBoost, and Gradient Boosting, sorted by Random Forest importance (descending).

NDVI, LAI, and NDRE are the three most informative features across all tree models. For Random Forest, NDVI (0.307), LAI (0.249), and NDRE (0.236) together account for 79% of importance; EVI contributes 0.112 and the remaining five features are near-zero. For XGBoost, LAI dominates strongly (0.646), with NDVI (0.183) and NDRE (0.134) distant second and third. Gradient Boosting shows a similar three-feature concentration: LAI (0.371), NDVI (0.306), NDRE (0.302). NDWI, BSI, and all three temporal NDVI statistics (ndvi_min, ndvi_max, ndvi_std) are collectively below 0.05 in RF importance, suggesting limited marginal discriminative power once the core spectral indices are included.

Caution on circularity: NDVI, NDRE, and EVI are the three indices that define the crop_stress_label via a weighted composite. Their high importance partly reflects the deterministic labelling rule rather than independent biological variation; the reported ranking therefore measures *spectral proxy separability*, not confirmed agronomic importance. LAI is a linear rescaling of EVI ($LAI = 3.618 \cdot EVI - 0.118$), so its apparent dominance in boosted models is likely an artefact of that transformation. The per-class F1 heatmap across all models is shown in Fig. 5.

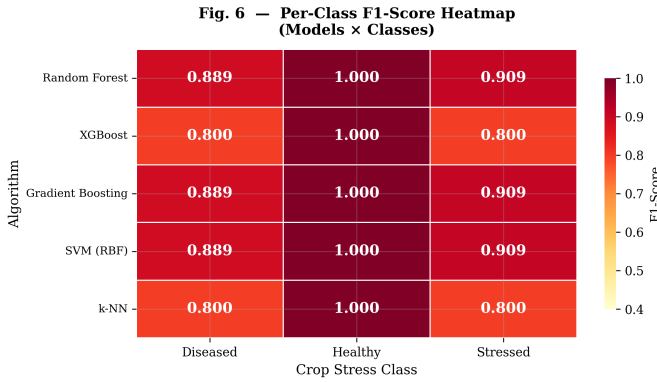


Fig. 5. Per-class F1-score heatmap across all five classifiers.

E. Discussion

The results demonstrate that nine Sentinel-2 spectral features — six derived indices and three temporal NDVI statistics — provide sufficient discriminative signal for spectral proxy crop-condition classification across Pakistan’s diverse agronomic zones. Macro F1-score was selected as the primary optimisation criterion to treat all three classes equally across their near-balanced support (596 Healthy, 595 Stressed, 595 Diseased). The location-grouped nested cross-validated macro F1 reached 0.9831 ± 0.0063 (XGBoost), 0.9813 ± 0.005 (Gradient Boosting), and 0.9804 ± 0.0066 (Random Forest) on the outer CV folds, with the held-out location-grouped test set yielding 0.9740 (XGBoost) and 0.9819 (SVM RBF).

Comparison with prior work requires a critical caveat: Htitiou *et al.* [13] reported 88% overall accuracy using Random Forest for crop-type mapping with ground-truth field surveys, and Maleki *et al.* [17] achieved F1 scores of 89.9–93.5% for

crop mapping using multi-sensor fusion and in-situ reference data. These works used independently verified class labels. By contrast, AgroSense labels are *spectrally derived* from the same NDVI, NDRE, and EVI values used as features; the comparison is therefore not directly commensurate. Our high F1 values partially reflect spectral proxy separability rather than independent agronomic validation. Future work with GPS-surveyed, pathology-confirmed field samples is needed for a fair cross-study comparison.

A key methodological contribution is the use of *location-grouped* train/test splitting: the dataset contains multiple seasonal observations per sampling location, and a naive random split would allow the same location to appear in both training and test partitions, inflating accuracy estimates. By applying `GroupShuffleSplit` on location identifiers, the held-out test set contains only sampling locations unseen during training, providing a more conservative and realistic estimate of generalisation to new locations. Similarly, nested cross-validation (5-fold outer / 3-fold inner `StratifiedGroupKFold`) was used to avoid overfitting hyperparameter selection to the validation split.

Reported metrics should be interpreted as spectral proxy classification performance. Class labels are derived from the same vegetation indices used as features; ground-truth field validation remains an important direction for future work to establish independently verified crop-condition detection accuracy.

VIII. API DESIGN

The backend exposes a RESTful API with JWT bearer authentication. The API follows resource-oriented design principles: each endpoint operates on a single identifiable resource (farm, field, or analysis result) and is stateless across requests, enabling independent horizontal scaling of the inference tier. Table VII summarises the primary endpoints.

IX. DEPLOYMENT

The backend is containerised via Docker and deployed to Google Cloud Run (us-central1), providing auto-scaling to zero for cost efficiency, a stable HTTPS endpoint, and eliminating infrastructure management requirements. The PostgreSQL database is hosted on a managed instance with the PostGIS 3 [24] extension. All third-party integrations (GEE, Open-Meteo, Gmail SMTP) are handled server-side; the mobile client holds only a short-lived, revocable JWT token with no embedded API keys.

X. DATA AND CODE AVAILABILITY

The dataset (1,786 Sentinel-2 observations), training and evaluation scripts, GEE data collection pipeline, Jupyter notebook, and pre-trained models are publicly available at:

<https://github.com/ALI-SHAHMEER/agrosense-pakistan-crop-stress>

The repository includes a data dictionary, full methodology documentation, and pinned dependency versions to support exact replication of the reported results.

TABLE VII
SUMMARY OF PRIMARY API ENDPOINTS

Endpoint	Method	Purpose
/auth/login	POST	Authenticate user; issue JWT access token
/auth/register	POST	Create new user account
/farms	GET / POST	List farms or create a new farm
/fields	GET / POST / DELETE	Field CRUD with PostGIS boundary polygon
/imagery/{field_id}	POST	Trigger GEE satellite fetch and index computation
/ml/full-analysis/{field_id}	GET	Run all five ML models and return combined report
/ml/crop-stress/{field_id}	GET	Crop stress classification (Healthy / Stressed / Diseased)
/ml/irrigation/{field_id}	GET	Irrigation recommendation (4-class output)
/ml/yield/{field_id}	GET	Yield prediction (t/ha) and harvest readiness
/ml/soil/{field_id}	GET	Soil health (pH, salinity dS/m, organic matter %)
/ml/vra-zones/{field_id}	GET	Variable-rate application zone mapping
/ml/export-pdf/{field_id}	GET	Generate and stream ReportLab PDF report
/ml/send-alert/{field_id}	POST	Dispatch crop health alert via Gmail SMTP
/weather/smart-farming/{field_id}	GET	7-day forecast, alert rules, and planting date scorer

XI. CONCLUSION

AgroSense demonstrates that satellite remote sensing, machine learning, and real-time weather intelligence can be integrated into an accessible mobile platform calibrated to the agronomic and linguistic context of Pakistan’s smallholder farming sector. Three findings emerge from this work. First, nine Sentinel-2 spectral features (six derived indices and three temporal NDVI statistics) provide sufficient discriminative signal for spectral proxy crop-condition classification across all four provinces of Pakistan, with SVM (RBF) achieving the highest location-grouped held-out test accuracy (98.34%) and XGBoost achieving the best nested cross-validated macro F1 (0.983 ± 0.006) — both under location-grouped evaluation. Reported performance measures spectral proxy separability; class labels are derived from the same spectral indices used as features, and independent field validation is future work. Second, location-grouped train/test separation (GroupShuffleSplit on location identifiers) is essential for honest accuracy estimation when datasets contain multiple seasonal observations per sampling location; naive random splits inflate reported accuracy by allowing the same location to appear in both partitions. Third, a single stateless REST API deployed on Google Cloud Run [8], [22] can serve both a Flutter mobile application [23] and a PyQt6 desktop application, enabling simultaneous support for field and office use cases without code duplication.

Future work will focus on ground-truth field validation of the spectrally derived class labels, incorporation of multi-temporal deep learning architectures, and formal user studies with smallholder farmers to evaluate decision quality under real field conditions.

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